



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Statistics for Data Analytics				
Course Code:	QMM1001	Credit Value:	4	Credit Hours:	56
Programs:	BAPG Business Analytics CAGC Crime Analytics HAGC Health Analytics				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

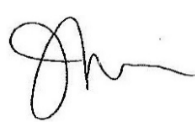
General Education Course: ☐ Degree Breadth Course: ☐ Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will investigate descriptive statistics, linear regression, probability concepts, inferential statistics and more foundational statistical tools that are applicable to data analytics. Students will also explore the R programming language for use in statistical computing and data visualizations.

Date: September 07, 2021

Prepared By: Sidney Shapiro

Approved by: 

JILL FERGUSON
Chair, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
Business Analytics	College Standards 1. Collect, manipulate and mine data sets to meet organizational need. 2. Design and apply data models that meet the needs of a specific operational/business process. 3. Design and present data visualizations to communicate information to business stakeholders. 4. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 5. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.
Crime Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Design and present data visualizations to communicate information to stakeholders.
Health Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Design health indicators and apply data models that meet organizational, program and service needs. 3. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 4. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Types of Variables
- Surveys and Sampling
- Describing and Displaying Categorical and Quantitative Data
- Correlation and Linear Regression
- Basic Probability
- Confidence Intervals
- Hypothesis Testing
- Utilizing R for Statistical Computing

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Describe core concepts in statistics.	1.1 Define what data and statistics are. 1.2 Identify whether a variable is categorical or quantitative. 1.3 Identify whether a variable is nominal, ordinal, interval, or ratio.
2. Explain core concepts of sampling theory.	2.1 Define the main ideas and terms associated with sampling. 2.2 Distinguish between simple random, stratified, cluster, systematic, and multistage samples. 2.3 Identify the principles in designing a valid survey. 2.4 Identify causes of bias and how to avoid them.
3. Construct and interpret visual displays of categorical data.	3.1 Create and interpret a frequency table. 3.2 Create and interpret bar charts and pie charts. 3.3 Create and interpret a contingency table.
4. Use various methods for describing and displaying quantitative data.	4.1 Create and interpret histograms, boxplots, and time series plots. 4.2 Describe the shape of a distribution (symmetric, skewed) and identify the mode(s) and outlier(s). 4.3 Calculate the mean and median and distinguish where each is appropriate and why. 4.4 Calculate the range, standard deviation, variance, and interquartile range, and distinguish where each is appropriate and why. 4.5 Calculate and interpret z-scores and use them to standardize values. 4.6 Use boxplots and z-scores to identify outliers.
5. Summarize a relationship using correlation and regression.	5.1 Create scatter plots and state the direction, form, and strength of the relationship they illustrate. 5.2 Check the three conditions required to use correlation. 5.3 Calculate the correlation coefficient r and illustrate its meaning and properties. 5.4 Distinguish correlation from causation. 5.5 Determine the regression line for a linear relationship and use it to interpolate/extrapolate data points.
6. State the basic principles of probability and use them to solve problems.	6.1 Know and apply the basic probability axioms. 6.2 State and apply the Law of Large Numbers. 6.3 Know the basic probability rules and apply them to solve problems.

Learning Outcome	Objectives
	6.4 Construct and interpret contingency tables. 6.5 Determine if events are independent.
7. Describe and perform calculations with discrete random variables.	7.1 State the basic definitions regarding discrete random variables. 7.2 Calculate and interpret the expected value of a discrete random variable. 7.3 Explain and solve problems regarding the binomial distribution and its parameters.
8. Determine probabilities for normally distributed data.	8.1 Identify basic properties of the normal distribution. 8.2 Define and calculate percentiles. 8.3 Perform probability calculations for normally distributed values with the aid of a standard normal distribution table.
9. Identify and solve problems involving sampling distributions.	9.1 Define and apply the concept of a sampling distribution. 9.2 Define the sampling distribution for the sample proportion. 9.3 Solve problems involving the sampling distribution for the sample proportion. 9.4 State and interpret the assumptions for the sampling distribution for proportions. 9.5 State and apply the Central Limit Theorem.
10. Construct, utilize, and interpret confidence intervals for proportions.	10.1 Construct, interpret, and solve problems involving confidence intervals for proportions. 10.2 Identify the relationship between sample size and a confidence interval. 10.3 Compute sample sizes for confidence intervals for proportions. 10.4 State and check the assumptions for confidence intervals.
11. Perform and interpret hypothesis testing for proportions.	11.1 Interpret and formulate the null and alternative hypotheses. 11.2 Perform a hypothesis test for a proportion. 11.3 Describe and interpret p-values and alpha-levels. 11.4 Correctly choose whether to perform a one-sided or two-sided hypothesis test. 11.5 Be able to identify Type I and Type II errors. 11.6 Illustrate the importance of effect size.
12. Perform basic statistical computations using the R programming language.	12.1 Create charts for categorical variables using R. 12.2 Describe and display quantitative data using R. 12.3 Create scatter plots, compute r and the regression line, and predict data points using R.

Learning Outcome	Objectives
	12.4 Compute normal probabilities with R. 12.5 Perform probabilistic simulations in R. 12.6 Determine confidence intervals using R. 12.7 Perform hypothesis tests using R.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- not applicable

Personal

- not applicable

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- eLearning Components

Resources Required

Books

Sharpe, De Veaux, Velleman & Wright, *Business Statistics*, 3rd, Pearson
ISBN: 0133899128

Norean R. Sharpe Richard D. De Veaux Paul F. Velleman David Wright David Wright, *Business Statistics, Fourth Canadian Edition, 4th edition*, 4th, Pearson
ISBN: 9780136964032

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
There will be multiple assignments in this course. Please see your instructor for details as the assignments may be in the form of written case studies, presentations, individual or group projects, or another form not listed.	
Applied Activities (G)	60%
There will be multiple computer labs requiring the use of the R programming language as determined by the instructor.	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)

Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>